Notes on the b-Calculus and the APS Index Theorem

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The argument is organized by dependency. We first isolate the failure of closed-manifold Fredholm theory on a cylinder. We then construct the b-double space, b-symbolic calculus, normal operator, weighted spaces, and full parametrix. Only after those tools are available do we return to the APS spectral realization and the two index proofs.

$$\begin{tikzcd}
\text{closed theory} \arrow[r] & \text{cylinder obstruction} \arrow[r] & \text{b-geometry} \arrow[d] \\
\text{APS index} & \text{full parametrix} \arrow[l] & \text{symbol + normal family} \arrow[l]
\end{tikzcd}$$

Figure 1: The principal symbol controls interior frequency; the normal operator controls escape along the cylinder. Their joint inverse produces the Fredholm parametrix used by the APS realization and the index proofs.

1 Elliptic Fredholm theory on closed manifolds

The b-calculus is a boundary refinement of ordinary elliptic theory, so we first record the closed-manifold argument that will later fail at the cylindrical end. Let M be closed and let

$$P : C^\infty(M; E) \longrightarrow C^\infty(M; F)$$

be a differential or classical pseudodifferential operator of order m. Its principal symbol

$$\sigma_m(P)(x, \xi) : E_x \longrightarrow F_x$$

is obtained by retaining the terms homogeneous of degree m in ξ. The operator is elliptic when this map is invertible for every $\xi \neq 0$.

For every $s \in \mathbb{R}$, an operator of order m extends boundedly as

$$P : H^s(M; E) \longrightarrow H^{s-m}(M; F).$$

Ellipticity permits the inverse symbol to be quantized and corrected recursively.

Theorem 1.1 (Elliptic parametrix). *If P is elliptic, there is $Q \in \Psi^{-m}(M; F, E)$ such that*

$$QP = I - R_L, \qquad PQ = I - R_R, \qquad R_L, R_R \in \Psi^{-\infty}(M).$$

Proof. Let $p \sim p_m + p_{m-1} + \cdots$ be the complete symbol of P. Choose $q_{-m} = p_m^{-1}$ away from the zero section. If $q_{-m}, \ldots, q_{-m-j+1}$ are known, the degree $-j$ part of $q\#p - I$ is affine in the unknown q_{-m-j}, with coefficient p_m. Multiplication by p_m^{-1} therefore determines q_{-m-j} and removes that term. Borel summation produces a classical symbol q for which $q\#p - I$ has order $-\infty$. Repeating on the other side and applying a smoothing correction gives one Q with both displayed identities. □

The logic of the construction is:

$$\begin{tikzcd} P \arrow[r, "\sigma_m"] & p_m \arrow[r, "\text{invert}"] & p_m^{-1} \arrow[d, "\text{symbol recursion}"] \\ \text{Fredholm} & I - \text{compact} \arrow[l, "\text{Atkinson}"'] & Q \arrow[l, "\text{compose}"'] \end{tikzcd}$$

Theorem 1.2 (Closed elliptic Fredholm theorem)**.** *For every s, the map*

$$P : H^s(M; E) \longrightarrow H^{s-m}(M; F)$$

is Fredholm. Its kernel is smooth, its range is closed, and

$$\operatorname{coker} P \cong \ker P^*.$$

If $Pu \in H^{r-m}$, then $u \in H^r$.

Proof. The parametrix gives

$$u = QPu + R_L u.$$

This proves elliptic regularity because Q gains m derivatives and R_L is smoothing. On a closed manifold, a smoothing operator factors through $H^{s+N} \hookrightarrow H^s$, which is compact by Rellich's lemma. Thus P is invertible modulo compact operators and is Fredholm by Atkinson's theorem. The orthogonal complement of its closed range is $\ker P^*$, giving the cokernel statement. □

The analytic index

$$\operatorname{ind} P = \dim \ker P - \dim \ker P^*$$

is independent of s and constant in continuous families of elliptic operators. A lower-order perturbation preserves the principal symbol and hence the index.

Lemma 1.3 (Fedosov trace formula)**.** *Suppose Q is a parametrix for the Fredholm operator P and*

$$S_L = I - QP, \qquad S_R = I - PQ$$

are trace class. Then

$$\operatorname{ind} P = \operatorname{Tr}(S_L) - \operatorname{Tr}(S_R).$$

Proof. Let G be the generalized inverse of P, equal to the inverse from $(\ker P)^\perp$ onto $\operatorname{ran} P$ and zero on $\ker P^*$. If Π_0, Π_1 are the orthogonal projections onto $\ker P, \ker P^*$, then

$$GP = I - \Pi_0, \qquad PG = I - \Pi_1.$$

Hence $\operatorname{ind} P = \operatorname{Tr}(\Pi_0) - \operatorname{Tr}(\Pi_1)$. Subtracting these identities from the identities for Q shows that $(Q - G)P$ and $P(Q - G)$ are trace class. Cyclicity gives them equal traces, so

$$\operatorname{Tr}(S_L) - \operatorname{Tr}(S_R) = \operatorname{Tr}(\Pi_0) - \operatorname{Tr}(\Pi_1).$$

□

Discussion. Both hypotheses in the closed theorem matter. Ellipticity produces smoothing remainders; compactness of the base turns the Sobolev gain into a compact map. On a cylinder, translations of a fixed bump remain bounded but have no convergent subsequence. The b-calculus must therefore control not only high frequency, through the principal symbol, but also escape to infinity, through the normal operator.

2 From the closed index theorem to the APS formula

Let X be a compact even-dimensional Riemannian manifold with boundary Y. Assume that the metric and all bundle data are of product type on a collar $[0,1)_u \times Y$. Let $E = E^+ \oplus E^-$ be a graded Hermitian Clifford module. Write the chiral operator as

$$P = D^+ : C^\infty(X; E^+) \longrightarrow C^\infty(X; E^-)$$

and the total odd self-adjoint operator as

$$\mathcal{D} = \begin{pmatrix} 0 & P^* \\ P & 0 \end{pmatrix}.$$

On the collar we use the convention

$$P = c(du)(\partial_u + A^+), \qquad \mathcal{D} = c(du)(\partial_u + \mathcal{A}),$$

where A^+ is self-adjoint on $E^+|_Y$ and

$$\mathcal{A} = \begin{pmatrix} A^+ & 0 \\ 0 & A^- \end{pmatrix}, \qquad A^- = -c(du)A^+c(du)^{-1}.$$

Thus $c(du)$ carries the λ-eigenspace of A^+ to the $-\lambda$-eigenspace of A^-. The eta invariant in the APS formula is always that of A^+, not that of the total block operator $\mathcal{A}$, whose two spectra cancel.

2.1 Closed manifolds

When $\partial X = \varnothing$, the heat operators are smoothing and trace class. Define

$$Q_t = P^* \int_0^t e^{-sPP^*}\, ds.$$

Functional calculus and the intertwining relation $P^* e^{-sPP^*} = e^{-sP^*P} P^*$ give

$$PQ_t = I - e^{-tPP^*}, \qquad Q_t P = I - e^{-tP^*P}.$$

Fedosov's trace formula therefore yields the McKean–Singer identity

$$\operatorname{ind} P = \operatorname{Tr}(e^{-tP^*P}) - \operatorname{Tr}(e^{-tPP^*}) = \operatorname{Str}(e^{-t\mathcal{D}^2})$$

for every $t > 0$. The local heat expansion then gives

$$\operatorname{ind} P = \int_X \left[\widehat{A}(TX, \nabla^{TX}) \operatorname{ch}(E/S, \nabla^{E/S}) \right]_{[n]}.$$

2.2 The cylindrical end

Attach $(-\infty, 0] \times Y$ to X and extend (P) and $\mathcal{D}$ by their product formulas. The resulting complete manifold is denoted X_∞. The operator

$$D^+ : H^1(X_\infty, E^+) \longrightarrow L^2(X_\infty, E^-)$$

is Fredholm precisely when A^+ is invertible. Expand a section in an eigenbasis of A^+. On the mode with eigenvalue λ, the operator on the end reduces to $\partial_u + \lambda$. A spectral gap gives uniform exponential estimates. If $0 \in \operatorname{spec}(A^+)$, translates of a fixed cutoff in a zero mode form a Weyl sequence at zero.

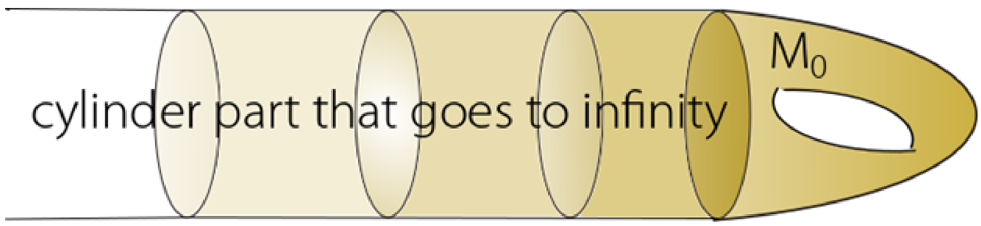


Figure 2: The infinite cylinder becomes a boundary hypersurface under $x = e^{-t}$. The original compact piece is on the right; translation along the attached cylinder becomes dilation toward $x = 0$.

Although D^+ is Fredholm when A^+ is invertible, the two heat operators are not separately trace class: their kernels are translation invariant to leading order on the infinite cylinder. Their graded difference is handled by the b-trace introduced below.

For $\Re s$ large, define the eta function

$$\eta_{A^+}(s) = \sum_{\lambda \in \operatorname{spec}(A^+) \setminus \{0\}} \operatorname{sgn}(\lambda) |\lambda|^{-s}.$$

Its Mellin representation is

$$\eta_{A^+}(s) = \frac{1}{\Gamma\left(\frac{s+1}{2}\right)} \int_0^\infty t^{(s-1)/2} \operatorname{Tr}(A^+ e^{-t(A^+)^2})\, dt.$$

The right side extends meromorphically and is regular at $s = 0$. With $h = \dim\ker A^+$, the Atiyah–Patodi–Singer formula is

$$\operatorname{ind} D^+_{\mathrm{APS}} = \int_X \left[\widehat{A}(TX)\,\mathrm{ch}(E/S)\right]_{[n]} - \frac{\eta_{A^+}(0) + h}{2}.$$

When A^+ is invertible, $h = 0$, and this equals the L^2 index on X_∞.

Question 2.1. *Why does attaching a cylinder help if the resulting manifold is noncompact?*

Answer. The cylinder does not remove the analytic difficulty; it puts that difficulty into a translation-invariant form. On the compact manifold with boundary, the APS condition is the nonlocal projection determined by the spectrum of A^+. On the cylinder, the same condition is encoded by square integrability. Indeed, on an A^+-eigenvector with eigenvalue λ, the equation $(\partial_u + A^+)u = 0$ becomes an elementary ODE, and exactly one spectral half gives decaying solutions on the outward end. Thus the APS projection and the L^2 condition select the same boundary data. The remaining translation-invariant part is what the normal operator and the b-trace measure. □

3 Real blow-up and the b-double space

A smooth n-manifold with corners is locally modelled on

$$\mathbb{R}^{n,k} = [0,\infty)^k \times \mathbb{R}^{n-k}, \qquad 0 \le k \le n,$$

with smooth transition maps in the usual extension sense. A boundary hypersurface is a connected codimension-one boundary face. At a point p, the codimension is the number of independent boundary defining functions that vanish at p.

Let $S \subset X$ be an embedded boundary face. Its real blow-up $[X;S]$ replaces S by the inward-pointing spherical normal bundle S^+NS. The blow-down map

$$\beta : [X;S] \longrightarrow X$$

is a diffeomorphism away from the new boundary hypersurface and collapses that hypersurface onto S.

For a compact manifold X with boundary, the b-double space is

$$X_b^2 = [X^2; \partial X \times \partial X].$$

Its boundary hypersurfaces are the front face ff and the lifts lb and rb of $\partial X \times X$ and $X \times \partial X$. The lifted diagonal Δ_b is the closure of the interior diagonal.

The geometry is summarized by the blow-down square

$$\begin{tikzcd} \Delta_b \arrow[r, hook] \arrow[d, "\beta|_{\Delta_b}"'] & X_b^2 \arrow[d, "\beta"] \\ \Delta \arrow[r, hook] & X^2. \end{tikzcd} \qquad \beta(\mathrm{ff}) = \partial X \times \partial X.$$

The blow-down is an isomorphism away from $\partial X \times \partial X$. Over that corner, the front face records the ratio of the two normal variables and therefore remembers how the corner is approached.

Near ∂X, choose product coordinates (x, y) and (x', y') on the two factors. Two useful projective coordinate systems are

$$(x, s = x'/x, y, y') \quad \text{and} \quad (s' = x/x', x', y, y').$$

In the first chart, $\mathrm{ff} = \{x = 0\}$ and $\mathrm{rb} = \{s = 0\}$; in the second, $\mathrm{ff} = \{x' = 0\}$ and $\mathrm{lb} = \{s' = 0\}$. Near the interior of the front face one may instead use

$$r = x, \qquad z = \log(x/x'), \qquad y, \qquad Y = y - y',$$

so that $\Delta_b = \{z = 0,\ Y = 0\}$.

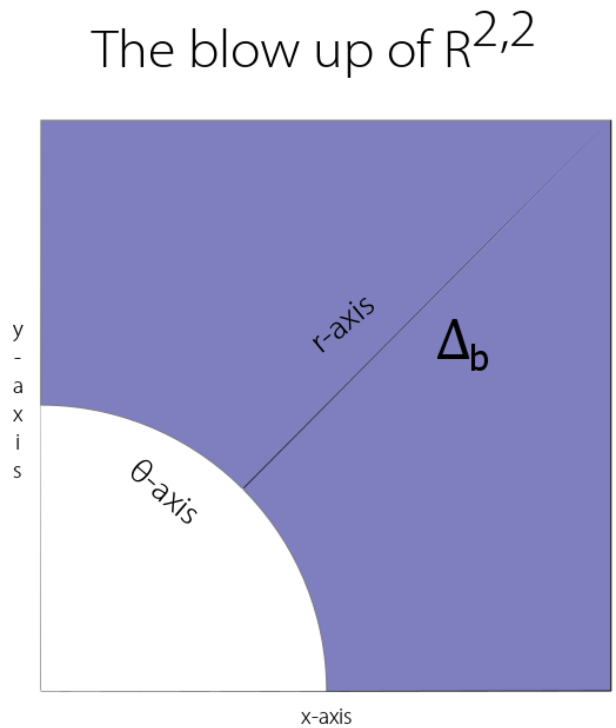


Figure 3: Blowing up the boundary corner replaces one point of approach by the quarter-sphere of inward normal directions. Projective coordinates on this front face retain the limiting ratio x/x'; the lifted diagonal meets it transversely.

Question 3.1. *Why not work directly on X^2?*

Answer. The ratio x/x' has no limit at $(x, x') = (0, 0)$ on X^2. Kernels of translation-invariant operators on the cylinder depend on $\log(x/x')$, so they cannot be smooth functions of x, x' at that corner. Blow-up replaces the corner by the set of limiting ratios. In the projective chart $x' = sx$, the same kernel is a function of $-\log s$, which is smooth in the interior of the front face and has controlled asymptotics as $s \to 0$ or $s \to \infty$. The singularity has not been discarded; it has been separated into geometrically distinct boundary faces. □

3.1 A one-dimensional model

On the full cylinder $\mathbb{R}_t$, consider

$$P = -\partial_t^2 + \epsilon^2, \qquad \epsilon > 0.$$

Near the end $t = -\infty$, use the compactifying coordinate $x = e^t$. Then

$$\partial_t = x\partial_x, \qquad dt = \frac{dx}{x}.$$

With respect to $dt' = dx'/x'$, the inverse of P has kernel

$$K_{P^{-1}}(x, x') = \frac{1}{2\epsilon} e^{-\epsilon|\log(x/x')|} = \frac{1}{2\epsilon} \begin{cases} (x'/x)^{\epsilon}, & x' \le x, \\ (x/x')^{\epsilon}, & x \le x'. \end{cases}$$

On X_b^2 this kernel is conormal to Δ_b, smooth across the front face away from Δ_b, and vanishes to order ϵ at lb and rb. This model explains why the b-calculus resolves the simultaneous limit $x, x' \to 0$ while retaining the translation-invariant variable $\log(x/x')$.

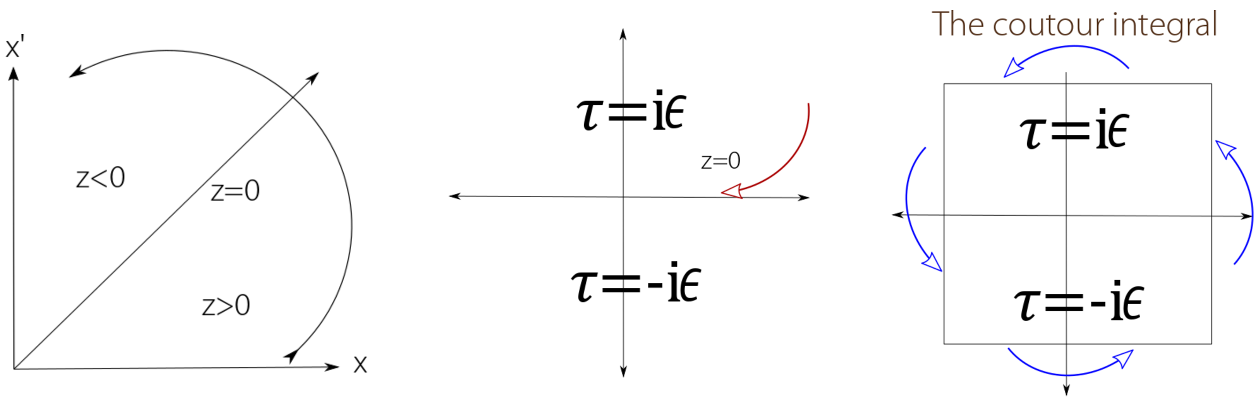


Figure 4: Left: the projective variable separates the regions $z < 0$, $z = 0$, and $z > 0$, where $z = \log(x/x')$. Center and right: the poles $\lambda = \pm i\epsilon$ determine which residue is enclosed when the Fourier–Mellin contour is closed. The two residues give the two branches of $e^{-\epsilon|z|}/(2\epsilon)$.

3.2 General and iterated blow-ups

The construction is not limited to a corner of a double space. A *p-submanifold* S of a manifold with corners Z is a submanifold that, in suitable product coordinates

$$(x_1, \dots, x_k, y_1, \dots, y_\ell),$$

is given by the vanishing of a subset of the x- and y-coordinates. Its inward spherical normal bundle is therefore well defined, and the real blow-up

$$[Z; S] = (Z \setminus S) \sqcup S^+NS$$

has a canonical smooth structure.

In the model $Z = [0, \infty)^2$, $S = \{(0, 0)\}$, the two projective charts are

$$U_x : (x, s = x'/x), \qquad U_{x'} : (s' = x/x', x').$$

On their overlap,

$$s' = s^{-1}, \qquad x' = sx.$$

These transition functions are smooth where $s, s' > 0$, and the front face is $x = 0$ in the first chart and $x' = 0$ in the second.

$$\begin{tikzcd} U_x = \{(x,s)\} \arrow[r, bend left, "x'=sx,\ s'=s^{-1}"] & U_{x'} = \{(s',x')\} \arrow[l, bend left, "x=s'x',\ s=(s')^{-1}"] \end{tikzcd}$$

Proposition 3.2 (Coordinate invariance)**.** *The blow-up $[Z;S]$, its front face, and its blow-down map are independent up to canonical diffeomorphism of the product coordinates used in the construction.*

Proof. A product-coordinate change preserving S has, in the normal variables, the form

$$\widetilde{z} = A(w)z + O(|z|^2),$$

where $A(w)$ is invertible and preserves the inward normal cone. Writing $z = r\omega$ gives

$$\widetilde{r} = r\,|A(w)\omega + O(r)|, \qquad \widetilde{\omega} = \frac{A(w)\omega + O(r)}{|A(w)\omega + O(r)|}.$$

Both expressions are smooth down to $r = 0$; the second restricts there to the induced map on the spherical normal bundle. Hence every allowed coordinate change lifts smoothly, and the lifted changes satisfy the same cocycle identities. □

Iterated blow-ups are used for X_b^3 and $X_{b,h}^2$. If two centers are disjoint or meet transversely, their blow-ups commute. If one center is contained in another, blowing up the smaller center first makes the lift of the larger one a p-submanifold. These facts justify the rule used below: blow up nested centers in increasing order of dimension and then blow up the transverse lifts.

Discussion. The order is geometric, not cosmetic. Blowing up a larger center first can destroy the product structure of a smaller center and make its lift ill-defined. The triple-space construction specifies the order precisely so that all three lifted projections become b-fibrations.

4 The b-tangent and b-cotangent bundles

Let $\mathcal{V}_b(X)$ be the Lie algebra of smooth vector fields tangent to every boundary face. In local coordinates

$$(x_1, \ldots, x_k, y_1, \ldots, y_{n-k}) \in [0,\infty)^k \times \mathbb{R}^{n-k},$$

it is the locally free $C^\infty(X)$-module generated by

$$x_1\partial_{x_1}, \ldots, x_k\partial_{x_k}, \partial_{y_1}, \ldots, \partial_{y_{n-k}}.$$

Consequently there is a vector bundle ${}^bTX \to X$ with

$$C^\infty(X; {}^bTX) = \mathcal{V}_b(X).$$

The natural anchor ${}^bTX \to TX$ is an isomorphism in the interior but drops rank at the boundary.

The dual bundle ${}^bT^*X$ has local frame

$$\frac{dx_1}{x_1}, \ldots, \frac{dx_k}{x_k}, dy_1, \ldots, dy_{n-k}.$$

A positive b-density is locally

$$a(x,y)\left|\frac{dx_1}{x_1}\cdots\frac{dx_k}{x_k}dy_1\cdots dy_{n-k}\right|, \qquad a > 0.$$

Under the cylindrical variables $x_j = e^{t_j}$, this is the ordinary product density $a\,|dt\,dy|$.

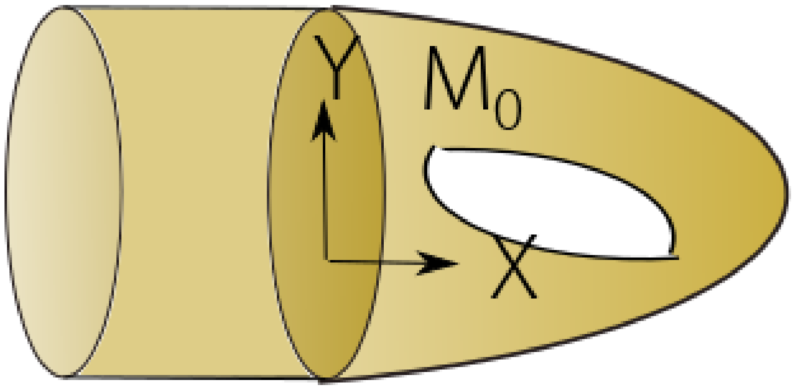


Figure 5: A b-vector field is tangent to the boundary. The normal generator $x\partial_x$ becomes the translation generator ∂_s for $s = -\log x$. It is nonzero in ${}^bTX$, although its anchor image in TX vanishes at $x = 0$.

5 b-differential operators and their symbols

The algebra $\mathrm{Diff}_b^m(X; E, F)$ consists of differential operators of order at most m generated over $C^\infty(X)$ by $\mathcal{V}_b(X)$. Locally,

$$P = \sum_{|\alpha|+|\beta|\leq m} a_{\alpha\beta}(x,y)(x\partial_x)^\alpha \partial_y^\beta.$$

For manifolds with corners, $(x\partial_x)^\alpha$ denotes $\prod_j (x_j\partial_{x_j})^{\alpha_j}$.

The filtration by order satisfies

$$[\mathrm{Diff}_b^m, \mathrm{Diff}_b^\ell] \subseteq \mathrm{Diff}_b^{m+\ell-1}.$$

It follows that the associated graded algebra is canonically $\mathrm{Sym}({}^bTX)$. Thus the principal symbol is the coordinate-independent homogeneous bundle map

$${}^b\sigma_m(P) : {}^bT^*X \setminus 0 \longrightarrow \mathrm{Hom}(\pi^*E, \pi^*F).$$

In the coordinates above,

$${}^b\sigma_m(P)(x,y;\xi,\eta) = \sum_{|\alpha|+|\beta|=m} a_{\alpha\beta}(x,y)(i\xi)^\alpha(i\eta)^\beta.$$

The symbol is multiplicative:

$${}^b\sigma_{m+\ell}(PQ) = {}^b\sigma_m(P)\,{}^b\sigma_\ell(Q),$$

and fits into the exact sequence

$$0 \longrightarrow \mathrm{Diff}_b^{m-1}(X;E,F) \longrightarrow \mathrm{Diff}_b^m(X;E,F) \xrightarrow{{}^b\sigma_m} \mathcal{P}^m({}^bT^*X; \mathrm{Hom}(\pi^*E,\pi^*F)) \longrightarrow 0.$$

Here $\mathcal{P}^m$ denotes smooth fibrewise homogeneous polynomials of degree m.

Definition 5.1. An operator $P \in \mathrm{Diff}_b^m(X; E, F)$ is b-elliptic if ${}^b\sigma_m(P)(\zeta)$ is invertible for every nonzero $\zeta \in {}^bT^*X$.

Fix a positive b-density and Hermitian metrics on the bundles. If $V \in \mathcal{V}_b(X)$, integration by parts gives

$$V^* = -V - \mathrm{div}_b(V).$$

Because $\mathrm{div}_b(V) \in C^\infty(X)$, formal adjoints preserve the calculus:

$$P \in \mathrm{Diff}_b^m(X; E, F) \quad \Longrightarrow \quad P^* \in \mathrm{Diff}_b^m(X; F, E).$$

This also gives

$${}^b\sigma_m(P^*) = {}^b\sigma_m(P)^*.$$

6 Conormal functions and boundary weights

Let X be a compact manifold with corners, with boundary hypersurfaces $H_1, \ldots, H_N$ and defining functions $\rho_1, \ldots, \rho_N$.

Definition 6.1. The space of bounded conormal functions is

$$\mathcal{A}^0(X) = \{u \in L^\infty(X^\circ) : V_1 \cdots V_k u \in L^\infty(X^\circ) \text{ for all } V_j \in \mathcal{V}_b(X)\}.$$

For a multiweight $\gamma = (\gamma_1, \ldots, \gamma_N)$, set

$$\rho^\gamma \mathcal{A}^0(X) = \rho_1^{\gamma_1} \cdots \rho_N^{\gamma_N} \mathcal{A}^0(X).$$

This definition is independent of the chosen defining functions. Indeed, a second defining function has the form $\widetilde{\rho}_j = a_j \rho_j$, where a_j is smooth and positive, and multiplication by $a_j^{\gamma_j}$ preserves $\mathcal{A}^0(X)$.

Conormality records uniform control under vector fields tangent to every boundary face. It is weaker than smoothness. Powers of defining functions belong to the corresponding weighted conormal classes. In contrast, $\log \rho_j \notin \mathcal{A}^0(X)$ because it is unbounded; logarithms belong only to broader logarithmic or suitably weighted conormal classes.

Definition 6.2. For $\epsilon > 0$, write

$$\mathcal{A}^{0,\epsilon}(X) = \mathcal{C}^\infty(X) + \rho^\epsilon \mathcal{A}^0(X), \qquad \rho^\epsilon = \prod_{j=1}^N \rho_j^\epsilon.$$

Thus $u \in \mathcal{A}^{0,\epsilon}(X)$ has a smooth boundary value and a conormal remainder that vanishes at every boundary hypersurface.

The same notation applies to sections of a smooth vector bundle after choosing local trivializations; the resulting space is independent of those trivializations.

7 Conormal distributions and the small b-calculus

Let $Z \subset M$ be an embedded submanifold of codimension k. In local coordinates (x, y), with $Z = \{x = 0\}$, a classical conormal distribution has the form

$$u(x, y) = (2\pi)^{-k} \int_{\mathbb{R}^k} e^{ix\cdot\xi} a(y, \xi)\, d\xi,$$

modulo a smooth term, where a is a classical symbol. The symbol order, with the standard dimension shift, defines the conormal order m and the space $I^m(M, Z)$. This description is invariant under coordinate changes.

For vector bundles $E, F \to X$, let ${}^b\Omega_R$ denote the lifted right b-density bundle on X_b^2. Set

$$\mathcal{H}_{E,F} = \mathrm{Hom}(E, F) \otimes {}^b\Omega_R.$$

Definition 7.1. The small b-pseudodifferential calculus is

$$\Psi_b^m(X; E, F) = \left\{ A : K_A \in I^m\left(X_b^2, \Delta_b; \mathcal{H}_{E,F}\right), \quad K_A \text{ vanishes to infinite order at } \mathrm{lb} \cup \mathrm{rb} \right\}.$$

The convention relating m to the conormal order depends on the density normalization; throughout these notes, m is the operator order. The lifted diagonal Δ_b meets the front face transversely, so restricting the kernel to the front face gives another conormal distribution.

Every b-differential operator of order m belongs to $\Psi_b^m(X)$: its Schwartz kernel is a finite sum of derivatives of the delta distribution on Δ_b. Operators in the small calculus act continuously

$$\Psi_b^m(X; E, F) : \mathcal{C}^\infty(X; E) \longrightarrow \mathcal{C}^\infty(X; F)$$

and preserve the space $\dot{\mathcal{C}}^\infty(X)$ of sections vanishing to infinite order at the boundary.

The principal symbol is a homogeneous bundle map on ${}^bT^*X \setminus 0$,

$${}^b\sigma_m(A) : \pi^*E \longrightarrow \pi^*F,$$

and fits into the exact sequence

$$0 \longrightarrow \Psi_b^{m-1}(X; E, F) \longrightarrow \Psi_b^m(X; E, F) \xrightarrow{{}^b\sigma_m} S^{[m]}({}^bT^*X; \mathrm{Hom}(E, F)) \longrightarrow 0.$$

Here $S^{[m]} = S^m / S^{m-1}$. An operator is b-elliptic when this symbol is invertible away from the zero section.

8 Composition in the b-calculus

Composition is most transparent on the b-triple space. Start with X^3 and blow up the triple corner and the lifted pairwise corners in increasing order of dimension. The resulting manifold with corners X_b^3 has b-fibrations

$$\pi_{12}, \pi_{23}, \pi_{13} : X_b^3 \longrightarrow X_b^2$$

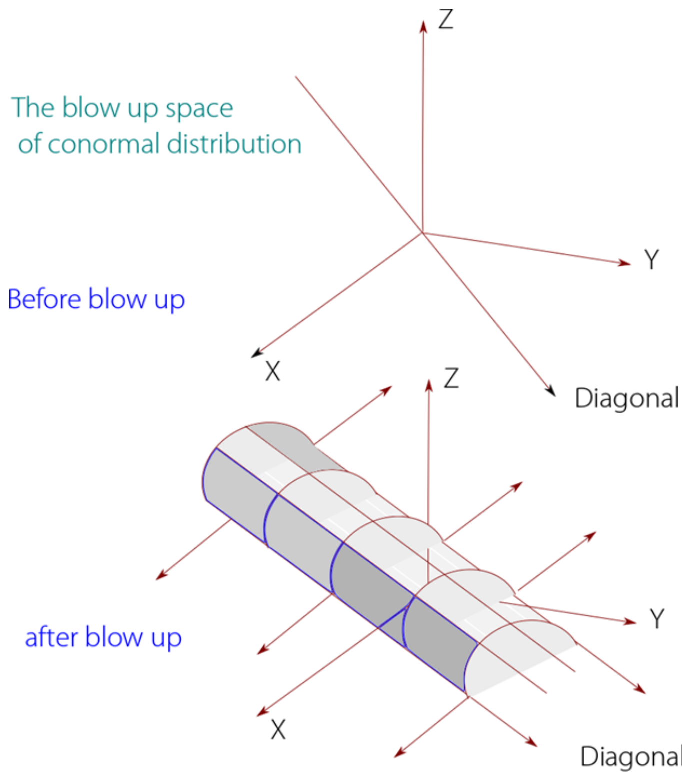


Figure 6: A small-calculus kernel is conormal to the lifted diagonal, represented by the sheet meeting the front face in the original diagram. It extends to ff and vanishes rapidly at the two side faces.

lifting the three coordinate projections.

Their roles in composition are displayed by

$$
\begin{tikzcd}
 & X_b^3 \arrow[dl, "\pi_{12}"'] \arrow[d, "\pi_{13}"] \arrow[dr, "\pi_{23}"] & \\
X_b^2 & X_b^2 & X_b^2.
\end{tikzcd}
$$

The left and right arrows pull back the two kernels; the middle arrow integrates out the shared variable.

Theorem 8.1 (Small b-composition). *If $A \in \Psi_b^m(X; E, F)$ and $B \in \Psi_b^{m'}(X; G, E)$, then*

$$K_{A\circ B} = (\pi_{13})_*\big(\pi_{12}^* K_A \; \pi_{23}^* K_B\big).$$

The lifted conormal sets meet cleanly, and the vanishing at the left and right faces makes the push-forward integrable. The pull-back, product, and push-forward theorems for conormal distributions therefore give

$$\Psi_b^m(X; E, F) \circ \Psi_b^{m'}(X; G, E) \subseteq \Psi_b^{m+m'}(X; G, F).$$

The leading oscillatory amplitudes multiply, so

$${}^b\sigma_{m+m'}(A \circ B) = {}^b\sigma_m(A)\, {}^b\sigma_{m'}(B).$$

Proof architecture; triple-space theorem quoted. The displayed push-forward formula follows from construction of the b-triple space. Its proof requires that the three projections be b-fibrations, clean pull-back and product of the lifted conormal distributions, and the push-forward theorem with the full index-family integrability conditions. These are the composition theorem of the b-calculus [3]; they are not consequences of ordinary Fubini alone.

To see the operator order once that theorem is available, work near the lifted diagonal and write the two kernels as oscillatory integrals. Their product has amplitude of order $m + m'$. Stationary phase in the intermediate variable identifies the two covectors and produces the usual symbol product. Away from the lifted diagonal, at least one factor is smooth; the push-forward is then smooth. At every boundary face, the exponent contributed by the product is the sum of the pulled-back exponents minus the fibre-density exponent. The infinite-order side-face vanishing in the small calculus places that sum strictly above the integrability threshold. □

The residual part of the full calculus allows polyhomogeneous expansions at the boundary faces. In the positive-weight case used below, products remain integrable and the index sets combine under the same pull-back and push-forward formula. Precise index-set bookkeeping is needed only when a leading exponent reaches an integrability threshold.

9 Parametrices in the small calculus

Let $P \in \Psi_b^m(X; E, F)$ be b-elliptic. Choose $Q_0 \in \Psi_b^{-m}(X; F, E)$ with

$$^b\sigma_{-m}(Q_0) = {}^b\sigma_m(P)^{-1}.$$

Then

$$PQ_0 = I - R_1, \qquad R_1 \in \Psi_b^{-1}(X; F).$$

The symbolic Neumann construction successively removes the homogeneous terms of R_1. Borel summation produces $Q \in \Psi_b^{-m}(X; F, E)$ such that

$$PQ = I - R_R, \qquad QP = I - R_L, \qquad R_R, R_L \in \Psi_b^{-\infty}(X).$$

This is a symbolic parametrix, not yet a Fredholm inverse. A residual kernel in the small b-calculus need not be compact on an unweighted b-Sobolev space: after passage to the cylindrical coordinate $t = -\log x$, it may retain translation-invariant behavior as $t \to \infty$. The normal operator, introduced next, detects precisely this missing boundary condition.

Question 9.1. *Why is a smooth residual kernel not automatically compact?*

Answer. Compactness on a compact manifold follows from smoothing plus Rellich's lemma. The b-metric turns a boundary collar into an infinite cylinder, so translations can escape every compact set. If a residual operator has a nonzero translation-invariant limit $K_\infty(t - t', y, y')$, choose a fixed compactly supported section u on the cylinder and translate it to $u_j(t, y) = u(t - j, y)$. The sequence u_j is bounded and has no convergent subsequence, while Ku_j is asymptotic to the corresponding translate of $K_\infty u$. Hence K is not compact unless its boundary model vanishes or decays. □

10 The normal operator and indicial family

The principal b-symbol controls high-frequency behavior, but it does not control escape along the cylindrical end. The missing invariant is the restriction of the lifted kernel to the front face.

Definition 10.1. For $A \in \Psi_b^m(X; E, F)$, the normal operator $N(A)$ is the translation-invariant operator on the inward normal cylinder of ∂X obtained by restricting the lifted kernel K_A to the front face.

After choosing a boundary defining function x and writing $t = -\log x$, the front face is modeled by $\partial X \times \partial X \times \mathbb{R}_t$. Thus $N(A)$ is a suspended pseudodifferential operator on ∂X. The normal map is multiplicative:

$$N(AB) = N(A)N(B).$$

It follows either from restriction of the composition formula to the front face or from Fourier transformation in t.

For a b-differential operator

$$P = \sum_{j+|\alpha|\le m} a_{j\alpha}(x,y)(x\partial_x)^j \partial_y^\alpha,$$

freezing the coefficients at $x=0$ gives

$$N(P) = \sum_{j+|\alpha|\le m} a_{j\alpha}(0,y)(-\partial_t)^j \partial_y^\alpha.$$

Definition 10.2. The indicial family of P is

$$I(P,\lambda) = x^{-i\lambda} P x^{i\lambda}\Big|_{\partial X} = \sum_{j+|\alpha|\le m} a_{j\alpha}(0,y)(i\lambda)^j \partial_y^\alpha.$$

For a b-pseudodifferential operator, $I(A,\lambda)$ is the Fourier transform of $N(A)$ in the translation variable, with

$$\widehat{f}(\lambda) = \int_{\mathbb{R}} e^{it\lambda} f(t)\,dt.$$

For a general small-calculus operator this is a suspended pseudodifferential family on the real λ-axis. It is not automatically an entire polynomial family: holomorphic continuation to a strip or sector requires the support and side-face estimates supplied by the relevant full calculus. Differential operators are the important exception above, for which the family is polynomial in λ.

If $x' = e^{\phi}x$, then

$$I_{x'}(A,\lambda) = e^{-i\lambda\phi_0} I_x(A,\lambda) e^{i\lambda\phi_0}, \qquad \phi_0 = \phi|_{\partial X}.$$

Thus the change is a λ-dependent conjugation by boundary multiplication operators, not a scalar unless ϕ_0 is constant. Invertibility is nevertheless intrinsic. Multiplicativity gives

$$I(AB,\lambda) = I(A,\lambda)I(B,\lambda).$$

The two symbol maps control complementary regimes:

$$\begin{tikzcd}
\Psi_b^m(X) \arrow[r, "{}^b\sigma_m"] \arrow[d, "N"'] & S^{[m]}({}^bT^*X) \\
\Psi_{\mathrm{sus}}^m(\partial X) \arrow[r, "\mathcal{F}_t"'] & \{I(A,\lambda)\}_{\lambda\in\mathbb{R}}.
\end{tikzcd}$$

The principal symbol sees large covectors at a fixed point. The indicial family sees bounded frequency while the base point escapes down the cylinder. Full ellipticity requires invertibility in both regimes.

Discussion. This is why the normal operator is often called a second symbol. It is not lower-order information: a smoothing b-operator has zero principal symbol but may have a nonzero normal operator, and that nonzero boundary model can destroy compactness.

For a product b-Dirac operator

$$D = c\left(\frac{dx}{x}\right)(x\partial_x + A_\partial),$$

one has

$$I(D,\lambda) = c\left(\frac{dx}{x}\right)(i\lambda + A_\partial).$$

Consequently D is invertible at the boundary on the unweighted line precisely when the tangential operator A_∂ is invertible.

11 Weighted b-Sobolev spaces

Fix a positive b-density and a positive b-Laplacian. For $s \in \mathbb{R}$, the b-Sobolev space $H^s_b(X;E)$ is defined by b-derivatives for integral s, and by the b-pseudodifferential calculus for arbitrary s. Near the boundary this is the ordinary Sobolev space in the variables $(t,y) = (-\log x, y)$.

For $\delta \in \mathbb{R}$, set

$$x^\delta H^s_b(X;E) = \{x^\delta u : u \in H^s_b(X;E)\}.$$

A b-operator of order m acts continuously as

$$P : x^\delta H^s_b(X;E) \longrightarrow x^\delta H^{s-m}_b(X;F).$$

Conjugation reduces this mapping to the unweighted one:

$$x^{-\delta} P x^\delta : H^s_b(X;E) \longrightarrow H^{s-m}_b(X;F).$$

With the convention $x^{i\lambda}$ used above, the weight x^δ corresponds to the Mellin line

$$\operatorname{Im}\lambda = -\delta.$$

Indeed, write $\lambda = \tau - i\delta$. Then $x^{i\lambda} = x^\delta x^{i\tau}$. Thus conjugating by x^δ moves the real Fourier line for the cylindrical variable to $\operatorname{Im}\lambda = -\delta$; no sign convention is hidden.

Definition 11.1. A b-elliptic operator P is fully elliptic at weight δ if

$$I(P,\lambda) \text{ is invertible whenever } \operatorname{Im}\lambda = -\delta.$$

For a pseudodifferential operator, this includes the uniform suspended parameter estimates for $I(P,\lambda)^{-1}$ on that line; pointwise invertibility without those estimates is not a mapping theorem.

For several boundary hypersurfaces, impose the corresponding condition at each face with a multiweight. Below, X has one boundary hypersurface.

12 Meromorphic inversion of the indicial family

In this section first suppose $P \in \mathrm{Diff}_b^m(X; E, F)$ is b-elliptic. Its indicial family is a polynomial holomorphic family of elliptic operators on the closed manifold ∂X. Parameter ellipticity gives Fredholm index zero and invertibility in suitable sectors for large modulus. Hence the family is invertible somewhere and the analytic Fredholm theorem applies. For a general b-pseudodifferential operator, every conclusion below remains valid only under the corresponding assumptions: a holomorphic suspended family, parameter ellipticity, index zero, and invertibility at one parameter.

Theorem 12.1 (Analytic Fredholm theorem)**.** *Let $T(\lambda)$ be a holomorphic family of Fredholm operators of index zero on a connected domain. If $T(\lambda_0)$ is invertible at one point, then $T(\lambda)^{-1}$ is meromorphic. Every coefficient in the principal part of a Laurent expansion at a pole has finite rank.*

Proof. Fix λ_*. Choose closed complements

$$X = \ker T(\lambda_*) \oplus X_1, \qquad Y = C \oplus \operatorname{ran} T(\lambda_*),$$

where C is finite-dimensional. Since the index is zero, $\dim C = \dim \ker T(\lambda_*)$. Relative to these decompositions, write

$$T(\lambda) = \begin{pmatrix} a(\lambda) & b(\lambda) \\ c(\lambda) & d(\lambda) \end{pmatrix}.$$

The block $d(\lambda_*) : X_1 \to \operatorname{ran} T(\lambda_*)$ is an isomorphism, so $d(\lambda)^{-1}$ is holomorphic near λ_*. Block Gaussian elimination reduces invertibility of $T(\lambda)$ to that of the finite-dimensional Schur complement

$$S(\lambda) = a(\lambda) - b(\lambda)d(\lambda)^{-1}c(\lambda) : \ker T(\lambda_*) \longrightarrow C.$$

Where S is invertible, the block inverse formula expresses $T(\lambda)^{-1}$ using $d(\lambda)^{-1}$ and $S(\lambda)^{-1}$. Cramer's rule gives

$$S(\lambda)^{-1} = \frac{\operatorname{adj} S(\lambda)}{\det S(\lambda)},$$

so this inverse is meromorphic. Its negative Laurent coefficients have range contained in a finite-dimensional space obtained from C and $\ker T(\lambda_*)$.

Join λ_0 to λ_* by a path and cover the path by finitely many such block neighborhoods $U_0, \ldots, U_N$, with connected nonempty successive overlaps and $\lambda_0 \in U_0$. On U_0, the determinant is not identically zero because $T(\lambda_0)$ is invertible. Inductively, suppose a meromorphic inverse has been constructed on U_j. Its poles are discrete, so $U_j \cap U_{j+1}$ contains a point where T is invertible. The Schur determinant on U_{j+1} is therefore not identically zero, and Cramer's rule constructs the next local meromorphic inverse. The two inverses agree on the nonempty open subset of the overlap where T is invertible, hence on the whole overlap by the identity theorem. This continues the inverse to U_N. Continuations along two paths agree by the same overlap argument (equivalently, by uniqueness in the meromorphic sheaf), so the local inverses assemble to one meromorphic inverse on the domain. $\square$

Applied to $I(P,\lambda)$, this theorem yields a discrete set

$$\mathrm{Spec}_b(P) = \{\lambda \in \mathbb{C} : I(P,\lambda) \text{ is not invertible}\},$$

called the boundary spectrum or set of indicial roots. The inverse $I(P,\lambda)^{-1}$ is meromorphic with finite-rank residues.

A weight is admissible exactly when its Mellin line avoids $\mathrm{Spec}_b(P)$. Moving the line without crossing an indicial root does not change the Fredholm index. When a root is crossed, the index jump is determined by its algebraic multiplicity; the sign depends on the direction in which the weight is moved.

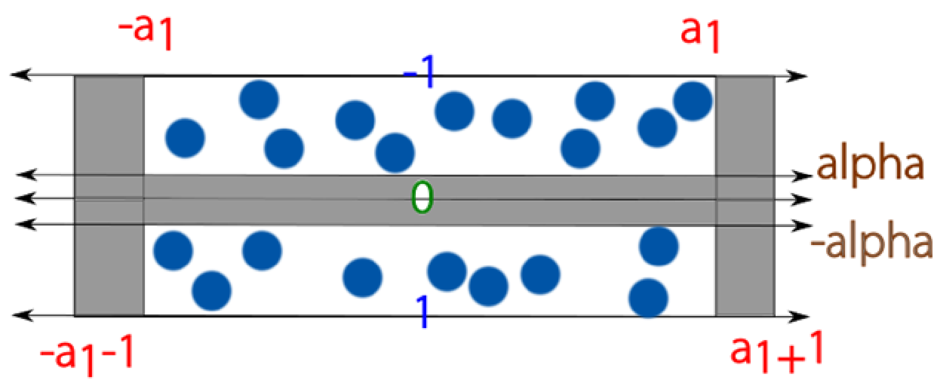


Figure 7: Changing the weight moves the Mellin contour. The index is constant while the contour remains in a pole-free strip and jumps by the algebraic multiplicities of the enclosed indicial roots when a pole is crossed.

13 The full b-calculus and the Fredholm parametrix

The symbolic parametrix constructed in the small calculus has smooth remainders, but their normal operators need not vanish. Full ellipticity allows one to remove this boundary error.

Theorem 13.1 (Full b-parametrix; quoted)**.** *Let P be b-elliptic and fully elliptic at the admissible weight δ, including the uniform parameter estimates in the preceding definition. Then the full b-calculus contains a two-sided parametrix Q such that*

$$PQ = I - R_R, \qquad QP = I - R_L.$$

The remainders are residual, have zero normal operators, gain arbitrarily many b-derivatives, and have strictly positive side-face orders after conjugation by x^δ. Their complete polyhomogeneous index families are obtained from the poles of $I(P,\lambda)^{-1}$.

This is the full parametrix theorem of the b-calculus [3, Chapters 4–5]. Its proof uses the extended union and composition of complete index families, rapid parameter estimates for inverse Mellin transformation, and Borel summation at every boundary face. The following calculation records its mechanism without replacing those theorems.

Let $Q_0 \in \Psi_b^{-m}$ be a symbolic parametrix for P:

$$PQ_0 = I - R_0, \qquad Q_0 P = I - R_0', \qquad R_0, R_0' \in \Psi_b^{-\infty}.$$

On an admissible Mellin line, define the boundary correction by

$$I(Q_1, \lambda) = I(Q_0, \lambda) + I(P, \lambda)^{-1} I(R_0, \lambda).$$

Then $I(PQ_1, \lambda) = I$. The inverse Mellin transform produces a kernel with polyhomogeneous expansions at the boundary faces. Its exponents are governed by the poles of $I(P, \lambda)^{-1}$. This is the reason the full b-calculus, rather than only the small calculus, is needed.

For the leading front-face correction, choose a horizontal line $\Gamma_\delta = \{\operatorname{Im} \lambda = -\delta\}$ avoiding the boundary spectrum. With cutoffs $\chi(x)\chi(x')$, the inverse Mellin formula

$$K_C(x, y; x', y') = \frac{1}{2\pi} \int_{\Gamma_\delta} \left(\frac{x}{x'}\right)^{i\lambda} I(P, \lambda)^{-1} I(R_0, \lambda)(y, y')\, d\lambda$$

defines a residual correction C near the front face. Parameter-elliptic estimates make the integral rapidly convergent in $\operatorname{Re} \lambda$. Applying P, using Mellin multiplicativity, and restricting to the front face gives

$$N(P(Q_0 + C) - I) = 0.$$

If the contour is shifted, the residue theorem adds terms

$$x^{i\lambda_0} (\log x)^j a_{\lambda_0, j}(y, y'),$$

where λ_0 is an indicial root and j is one less than the order of the corresponding pole. These are precisely the polyhomogeneous terms allowed in the full calculus.

At the level of complete index families, the construction is performed on both sides. Any mismatch between the two corrections is residual and has zero normal operator. A Neumann iteration in the residual ideal removes its successive boundary coefficients; Borel summation removes all of them at once. An interior symbolic correction then restores infinite-order vanishing at the lifted diagonal. The quoted full-calculus theorem justifies these iterations and yields the two-sided parametrix below.

Iterating at the interior diagonal and at the boundary faces gives $Q \in \Psi_b^{-m}$ in the full calculus with

$$PQ = I - R_R, \qquad QP = I - R_L.$$

At an admissible weight, R_R and R_L are residual operators whose normal operators vanish and whose side-face exponents have the decay required by that weight.

Lemma 13.2. *The remainders R_R and R_L act compactly on weighted b-Sobolev spaces.*

Proof. A residual kernel gains arbitrarily many b-derivatives. Vanishing of its normal operator removes the translation-invariant limit on the cylindrical end, while its positive side-face weights give uniform decay there. Cut the cylinder at $t = T$. On the compact

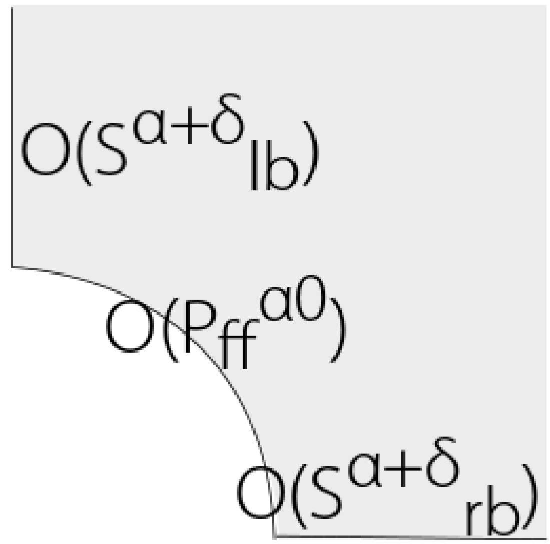


Figure 8: The full parametrix alternates interior symbol inversion with indicial correction at the front face. The face labels in the original diagram record the resulting orders at lb, ff, and rb; positivity at the side faces is what makes the final residual error compact.

part, Rellich's lemma makes the operator compact; on the tail, the operator norm tends to zero as $T \to \infty$. Hence the full operator is a norm limit of compact operators.

More explicitly, let χ_T equal one on $t \leq T$ and vanish on $t \geq T+1$. The operator $\chi_T R \chi_T$ factors as

$$H_b^s \xrightarrow{R} H_b^{s+N} \xrightarrow{\chi_T} H_b^s$$

for any $N > 0$; the second arrow is compact by Rellich on the truncated cylinder. The full-calculus residual mapping theorem, using both $N(R) = 0$ and the strictly positive side-face orders stated above, gives

$$\|(1-\chi_T)R\| + \|R(1-\chi_T)\| \longrightarrow 0.$$

Therefore

$$R - \chi_T R \chi_T = (1-\chi_T)R + \chi_T R(1-\chi_T)$$

tends to zero in operator norm. This proves compactness without assuming that the noncompact b-manifold itself has a compact Sobolev embedding. $\square$

Theorem 13.3 (Fredholm theorem for b-operators)**.** *Let $P \in \Psi_b^m(X; E, F)$ be b-elliptic and fully elliptic at weight δ. Then, for every $s \in \mathbb{R}$,*

$$P : x^\delta H_b^s(X; E) \longrightarrow x^\delta H_b^{s-m}(X; F)$$

is Fredholm.

Proof. The full b-parametrix Q is bounded between the reverse weighted Sobolev spaces, and

$$PQ = I - R_R, \qquad QP = I - R_L$$

with compact remainders. Atkinson's theorem therefore implies that P is Fredholm. $\square$

Corollary 13.4 (Elliptic and polyhomogeneous regularity). *If $Pu = f$, where $u \in x^\delta H_b^{s_0}$ and $f \in x^\delta H_b^{s-m}$, then $u \in x^\delta H_b^s$. If, in addition, the coefficients of P and f are polyhomogeneous with compatible index families, then u is polyhomogeneous. Its exponents are generated by those index families and the indicial roots; higher-order poles produce logarithms. In particular, a null-solution of a polyhomogeneous fully elliptic operator has such an expansion.*

Proof. Apply the parametrix:

$$u = Qf + R_L u.$$

The first term gains m derivatives, and the residual term gains arbitrarily many. Iteration gives the Sobolev regularity. The additional polyhomogeneous statement is the full-calculus regularity theorem: Mellin inversion and contour shifting apply because the coefficients, right-hand side, and parametrix have compatible complete index families. Residues at indicial poles then produce powers of x and, at higher-order poles, logarithmic terms. Sobolev iteration alone would not imply this expansion. □

14 The Dirac case

For the chiral operator

$$P = D^+ = c\left(\frac{dx}{x}\right)(x\partial_x + A^+),$$

the b-principal symbol is Clifford multiplication and is invertible away from the zero section. Thus P is b-elliptic. Its indicial family is invertible on the unweighted line if and only if A^+ is invertible.

If A^+ has kernel, one may choose a small nonzero weight δ whose Mellin line avoids the boundary spectrum. The resulting weighted realization is Fredholm, but its index depends on the side from which the zero indicial roots are crossed. When the contour crosses an indicial root, the weighted Fredholm index jumps by the full algebraic multiplicity of that root, with sign determined by the direction of the crossing.

This integer jump is distinct from the half-kernel term in the APS formula. The latter belongs to the reduced eta invariant

$$\bar{\eta}(A^+) = \frac{\eta(A^+) + \dim\ker A^+}{2}$$

and reflects the strict/non-strict choice at the zero eigenspace. It is not a weighted-index jump.

15 The APS boundary realization

The preceding b-Fredholm theorem explains the cylindrical realization. We now return to the compact manifold with boundary and prove that its spectral boundary condition selects exactly the same modes. Placing this section here avoids using the indicial family before it has been defined.

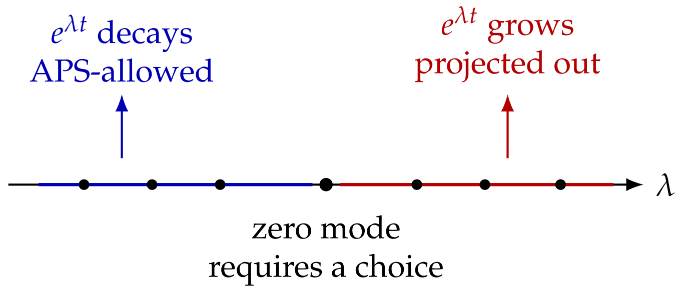


Figure 9: On the outward cylinder, the tangential spectrum splits boundary data into decaying and growing modes. The APS projection makes this split at $\lambda = 0$.

Let

$$\gamma : H^1(X; E^+) \longrightarrow H^{1/2}(Y; E^+|_Y)$$

be the trace map. Since A^+ is self-adjoint and elliptic on the closed manifold Y, it has discrete real spectrum, and its spectral projections are order-zero pseudodifferential operators. Define

$$\Pi_{\geq 0}(A^+) = \sum_{\lambda \geq 0} \Pi_\lambda$$

and

$$\mathcal{D}_{\mathrm{APS}}(D^+) = \left\{ u \in H^1(X; E^+) : \Pi_{\geq 0}(A^+)\gamma u = 0 \right\}.$$

Thus the boundary value is allowed to lie only in the negative spectral subspace of A^+. Since $\Pi_{\geq 0}(A^+) : H^{1/2}(Y) \to H^{1/2}(Y)$ and the trace map are bounded, this domain is the kernel of a bounded map from $H^1(X; E^+)$ to $H^{1/2}(Y; E^+|_Y)$. It is therefore closed in the H^1 graph norm.

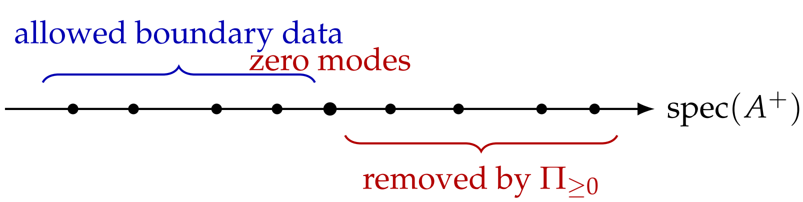


Figure 10: The APS condition is nonlocal: it uses the global spectral decomposition of the tangential operator rather than pointwise boundary values.

Green's formula determines the adjoint domain. With inward collar coordinate r,

$$\langle D^+ u, v \rangle_X - \langle u, D^- v \rangle_X = -\langle c(dr)\gamma u, \gamma v \rangle_Y.$$

Recall that $A^- = -c(dr)A^+c(dr)^{-1}$ is the tangential operator on $E^-|_Y$. Because Clifford multiplication sends the λ-space of A^+ to the $-\lambda$-space of A^-, Green's formula gives the explicit adjoint domain

$$\mathcal{D}((D^+_{\mathrm{APS}})^*) = \{ v \in H^1(X; E^-) : \Pi_{>0}(A^-)\gamma v = 0 \}.$$

Indeed, this condition makes γv orthogonal to $c(dr)\gamma u$ for every allowed negative boundary value of u, and maximality follows from surjectivity of the trace map. The strict projection at zero is the source of the kernel-dimension correction.

Theorem 15.1 (Fredholmness of the APS realization)**.** *The operator*

$$D^+_{\mathrm{APS}} : \mathcal{D}_{\mathrm{APS}}(D^+) \longrightarrow L^2(X; E^-)$$

is Fredholm. If A^+ is invertible, restriction to X and extension along the cylinder identify its kernel and cokernel with those of the L^2 realization on X_∞. Consequently

$$\operatorname{ind} D^+_{\mathrm{APS}} = \operatorname{ind}_{L^2} D^+_{X_\infty}.$$

Proof. Choose an orthonormal eigenbasis $A^+\phi_\lambda = \lambda\phi_\lambda$ and expand on the collar,

$$u(r, y) = \sum_\lambda u_\lambda(r)\phi_\lambda(y).$$

The product operator is the direct sum of the one-dimensional operators $\partial_r + \lambda$. For a scalar function f, integration by parts gives

$$\int_0^\varepsilon |f' + \lambda f|^2 dr = \int_0^\varepsilon (|f'|^2 + \lambda^2|f|^2)dr + \lambda|f(\varepsilon)|^2 - \lambda|f(0)|^2.$$

For $\lambda \geq 0$, the APS condition sets $f(0) = 0$. For $\lambda < 0$, the last term has the favorable sign. A cutoff at $r = \varepsilon$, followed by summation over λ, yields the collar estimate below. Here one uses

$$\|w\|^2_{H^1(Y)} \asymp \|A^+w\|^2_{L^2(Y)} + \|w\|^2_{L^2(Y)},$$

the elliptic graph-norm equivalence for A^+; the finitely many eigenvalues in a fixed bounded interval are absorbed by the L^2-term. Thus

$$\|u\|_{H^1([0,\varepsilon)\times Y)} \leq C\left(\|D^+u\|_{L^2} + \|u\|_{L^2} + \|u\|_{H^1(\{r>\varepsilon/2\})}\right).$$

Choose a partition of unity with one function supported in the collar and one in the interior. The commutators with these cutoffs are order zero. The ordinary interior elliptic estimate controls the interior function, and its compact lower-order remainder is absorbed into the L^2-term. This gives

$$\|u\|_{H^1(X)} \leq C\big(\|D^+u\|_{L^2(X)} + \|u\|_{L^2(X)}\big).$$

The same estimate holds for the adjoint problem just displayed. Rellich compactness first makes each kernel finite-dimensional: the unit ball of a kernel is precompact in L^2, and the estimate upgrades L^2 convergence to H^1 convergence.

For closed range, the estimate implies the sharper bound

$$\|u\|_{H^1} \leq C\|D^+u\|_{L^2}, \qquad u \perp \ker D^+_{\mathrm{APS}}.$$

Otherwise there would be $u_j \perp \ker D^+_{\mathrm{APS}}$ with $\|u_j\|_{H^1} = 1$ and $D^+u_j \to 0$. Rellich gives an L^2-convergent subsequence; applying the original estimate to differences makes it H^1-convergent to a unit vector in the kernel, contradicting orthogonality. The sharper bound makes the range closed. Repeating the argument for the adjoint gives finite-dimensional cokernel. Thus the APS realization is Fredholm.

Assume now that A^+ is invertible and write the outward cylinder coordinate as $t = -r \geq 0$. A solution in the λ-eigenspace is

$$u_\lambda(t) = e^{\lambda t} u_\lambda(0).$$

It is square integrable exactly for $\lambda < 0$, precisely the boundary data allowed by the APS projection. Restriction and extension identify the kernels. The complementary projection gives the same identification for the adjoints, hence for the cokernels. □

Question 15.2. *Why is an elliptic differential operator on a manifold with boundary not automatically Fredholm?*

Answer. The interior principal symbol controls high-frequency behavior away from the boundary but does not select boundary data. For example, $\bar{\partial}$ on a disk has the infinite-dimensional space of holomorphic functions in its kernel if no boundary condition is imposed. An elliptic boundary condition removes half of the Cauchy data. The APS condition makes that choice spectrally; the b-calculus encodes the same choice as decay on the attached cylinder. □

16 The K-theoretic proof architecture

The closed index theorem has two layers: an elliptic symbol defines a K-class, and the analytic index is the unique push-forward of that class normalized by the Bott operator. On a manifold with boundary, the interior symbol alone is insufficient. One must also record an invertible model at the cylindrical end. The b-calculus packages exactly these two pieces [3, 5].

Fix the full order-zero b-calculus on $L^2_b(X)$, stabilize by compact matrices, and let $\mathcal{A}_b$ be its operator-norm closure. Let Σ_b be the norm closure of compatible pairs in the product of the continuous b-cosphere symbols and the order-zero suspended normal algebra. An element of Σ_b is a compatible pair

$$(a, q) = \left({}^b\sigma_0(P), I(P, \lambda)\right),$$

where a is the b-principal symbol and q is the suspended indicial family. Compatibility means that the principal symbol of q is the restriction of a to the boundary.

Theorem 16.1 (Joint symbol extension; quoted)**.** *With these closures, the joint symbol map is surjective and the sequence*

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{A}_b \xrightarrow{j} \Sigma_b \longrightarrow 0$$

is exact.

This is a C*-closure theorem, including surjectivity of the compatibility map and identification of its kernel; it does not follow from the formal parametrix alone. We use the standard result of Melrose and Melrose–Nistor [3, 4]. A b-operator with vanishing principal and normal symbols lies in the compact kernel, while the theorem supplies the converse and the closure assertions.

If P is fully elliptic, $j(P)$ is invertible. After stabilization it defines a class $[j(P)] \in K_1(\Sigma_b)$. The six-term exact sequence has a connecting map

$$\partial : K_1(\Sigma_b) \longrightarrow K_0(\mathcal{K}) \cong \mathbb{Z}.$$

Lemma 16.2 (The connecting map is the Fredholm index)**.** *With the usual sign convention for ∂,*

$$\partial[j(P)] = \operatorname{ind} P.$$

Proof. Lift $j(P)^{-1}$ to a b-parametrix Q. Then

$$I - QP \in \mathcal{K}, \qquad I - PQ \in \mathcal{K}.$$

After a compact perturbation, the polar part V of P is a partial isometry. The definition of the connecting map gives

$$\partial[j(P)] = [I - V^*V] - [I - VV^*].$$

The first defect projection is the orthogonal projection onto $\ker P$, and the second is the projection onto $\ker P^*$. Under $K_0(\mathcal{K}) \cong \mathbb{Z}$, their difference is $\dim\ker P - \dim\ker P^* = \operatorname{ind} P$. □

For the chiral product operator $P = D^+$, the unbounded joint elliptic data are

$$\left({}^b\sigma_1(P),\, I(P,\lambda) = c\left(\frac{dx}{x}\right)(i\lambda + A^+)\right).$$

Composing on the right with the inverse of a fixed positive elliptic family of order one normalizes this to an order-zero element of Σ_b without changing invertibility or the index class. Equivalently, one may use the bounded transform of P, whose indicial family includes the corresponding factor of order minus one. If A^+ is invertible, this pair is fully elliptic. If A^+ has kernel, changing only the displayed boundary family does not yet define an operator on the cylinder. Put

$$V = \ker A^+, \qquad T = -I_V, \qquad \Pi_0^+ = \Pi_V.$$

The involution $T = -I_V$ is Loya's ideal boundary condition corresponding to the APS convention $\Pi_{\geq 0}(A^+)\gamma u = 0$: it moves every zero mode to the positive side. Choose a nonnegative even Schwartz function ρ with $\rho(0) = 1$. Surjectivity of the residual normal map supplies global lifts of the suspended family below; choose the Loya–Melrose–Piazza lift that realizes the ideal boundary condition. Thus there is a global chiral b-smoothing operator $T_b^+ : E^+ \to E^-$ with

$$I(T_b^+, \lambda) = c\left(\frac{dx}{x}\right)\varepsilon\rho(\lambda)\Pi_0^+.$$

For sufficiently small $\varepsilon > 0$, $i\lambda + A^+ + \varepsilon\rho(\lambda)\Pi_0^+$ is invertible for every real λ: on V it is the scalar $i\lambda + \varepsilon\rho(\lambda)$, which cannot vanish, and on $V^\perp$ the nonzero spectrum retains its gap. Hence

$$P_T = P + T_b^+$$

is fully elliptic. The associated total self-adjoint operator used in the heat proof is

$$\mathcal{D}_T = \begin{pmatrix} 0 & P_T^* \\ P_T & 0 \end{pmatrix}.$$

The perturbation is smoothing in the b-calculus, but it is not a finite-rank or compact operator on the infinite cylinder.

For this chosen lift, the ideal-boundary comparison theorem identifies the perturbed L^2 realization with the APS realization determined by $T = -I_V$: their kernels, adjoint kernels, and indices agree [2, 5]. A different lift with the same normal family has the same index, but need not have canonically identical kernel spaces. Thus the precise stable joint-symbol class is

$$[j(P_T)] \in K_1(\Sigma_b).$$

For the geometric push-forward it is safer to retain the relative symbol group rather than claim a Thom map on every element of $K_1(\Sigma_b)$. Let $\mathfrak{S}_b$ be the stabilized continuous b-cosphere-symbol algebra, let $\mathfrak{S}_\partial$ be its boundary restriction, and let $r : \mathfrak{S}_b \to \mathfrak{S}_\partial$ be restriction. Its mapping cone is

$$C_r = \{(a, f) \in \mathfrak{S}_b \oplus C_0([0,1), \mathfrak{S}_\partial) : f(0) = r(a)\}.$$

Write

$$K^0_{\mathrm{APS}}({}^bT^*X) := K_0(C_r).$$

The invertible indicial family of P_T is a boundary trivialization of its b-symbol and hence determines $\alpha_{\mathrm{APS}}(P_T) \in K^0_{\mathrm{APS}}({}^bT^*X)$. The comparison theorem for the joint-symbol pullback algebra supplies a map

$$\kappa : K^0_{\mathrm{APS}}({}^bT^*X) \longrightarrow K_1(\Sigma_b), \qquad \kappa(\alpha_{\mathrm{APS}}(P_T)) = [j(P_T)].$$

This mapping-cone/joint-symbol comparison is a quoted part of the relative index theorem [5]; it is not inferred from the formal parametrix.

A neat embedding

$$i : (X, \partial X) \hookrightarrow (\mathbb{R}^N_+, \mathbb{R}^{N-1})$$

defines a Thom push-forward of these relative symbol classes. Normal deformation and collar doubling identify the target with the Bott group

$$K^0_c(\mathbb{R}^{2N}) \cong \mathbb{Z}.$$

Call the resulting integer $\text{t-ind}_{\mathrm{APS}}$.

Theorem 16.3 (K-theoretic APS index theorem; quoted)**.** *For a product-type graded Dirac operator,*

$$\mathrm{ind}(D^+_{\mathrm{APS}}) = \text{t-ind}_{\mathrm{APS}}\big(\alpha_{\mathrm{APS}}(P_T)\big).$$

The theorem asserts that the following diagram commutes; the left vertical map is the connecting-map index after comparison with the joint symbol.

$$\begin{tikzcd} K^0_{\mathrm{APS}}({}^bT^*X) \arrow[r, "i_!"] \arrow[d, "\partial\circ\kappa"'] & K^0_c(\mathbb{R}^{2N}) \arrow[d, "\mathrm{Bott}"] \\ \mathbb{Z} \arrow[r, equal] & \mathbb{Z} \end{tikzcd}$$

Proof architecture. The proof follows the closed push-forward argument. In a tubular neighborhood, the Thom class is represented analytically by the fibrewise Bott operator

$$B_\nu = \sum_j c(e_j)\partial_{v_j} + \widehat{c}(v),$$

on $\Lambda^\bullet \nu^*_{\mathbb{C}}$, where $c(v) = \varepsilon(v^\flat) - \iota_v$ and $\widehat{c}(v) = \varepsilon(v^\flat) + \iota_v$. Its even kernel is the Gaussian line and its odd kernel is zero. Taking the graded product with D^+_{APS} therefore preserves the index. On the boundary collar the normal Bott factor is product type, so it carries the chosen indicial trivialization to the pushed-forward operator without changing which zero modes are selected. Excision removes the complement of the tubular neighborhood.

The existence and homotopy invariance of this relative Thom map, its compatibility with the joint-symbol extension, and the collar-doubling excision step are the K-theoretic APS push-forward theorem of Melrose and Melrose–Piazza [3, 5]. They are substantial inputs, not consequences of the displayed Bott formula alone.

After the neat embedding and collar doubling, the class is an integer multiple of the Bott generator. The Bott operator has index one, so both the analytic index, computed by the connecting map above, and the topological push-forward equal that integer. This proves the formula. □

This theorem proves the equality of two *integer* indices. Ordinary K-theory does not by itself determine the real number

$$\bar{\eta}(A^+) = \frac{\eta(A^+) + \dim\ker A^+}{2}.$$

That number is a secondary, differential refinement of the boundary K-class: it can vary continuously with the metric and connection even when the underlying K-class is fixed. The heat-kernel and b-trace argument below supplies its canonical real lift and proves the numerical formula

$$\mathrm{ind}(D^+_{\mathrm{APS}}) = \int_X \left[\widehat{A}(TX)\,\mathrm{ch}(E/S)\right]_{[n]} - \bar{\eta}(A^+).$$

Equivalently,

$$\bar{\eta}(A^+) \equiv \int_X \left[\widehat{A}(TX)\,\mathrm{ch}(E/S)\right]_{[n]} \pmod{\mathbb{Z}}.$$

Thus the exponentiated eta invariant is the boundary transgression of the local index form. The K-theoretic proof explains integrality and homotopy invariance, while the heat proof identifies the precise spectral correction.

Question 16.4. *Why can $\bar{\eta}(A^+)$ vary while the K-theoretic APS index is constant?*

Answer. As the geometric data vary without a zero eigenvalue crossing, the local integral and $\bar{\eta}(A^+)$ vary by the same amount, so their difference remains the fixed integer index. At a crossing, the spectral projection changes by a finite-dimensional space; spectral flow changes the K-theoretic boundary trivialization and accounts for the integer jump. Ordinary K-theory records that jump, whereas the eta invariant records the continuous real-valued refinement between jumps. □

17 The b-trace defect

We postpone regularization until this point because the b-density, front-face restriction, and indicial family are now available. Put $x = e^{-t}$ on the cylindrical end. If

$$f(x, y) = f_0(y) + O(x^\varepsilon)$$

with the same estimate after b-differentiation, then for $\operatorname{Re} z > 0$

$$\int_X x^z f \, d\mu_b = \frac{1}{z} \int_{\partial X} f_0 \, d\mu_{\partial X} + G_f(z),$$

where G_f is holomorphic for $\operatorname{Re} z > -\varepsilon$. Indeed, the leading term contributes $\int_0^1 x^{z-1} dx = z^{-1}$, and the remainder is integrable in the larger half-plane.

Definition 17.1. The b-integral is the finite part

$$^b\!\!\int_X f \, d\mu_b := \operatorname*{FP}_{z=0} \int_X x^z f \, d\mu_b = G_f(0).$$

Equivalently, if the cylinder is truncated at $t = T$, then

$$\int_{\{t \le T\}} f \, d\mu_b = cT + C + o(1), \qquad ^b\!\!\int_X f \, d\mu_b = C.$$

More generally, if the diagonal is a polyhomogeneous b-density

$$\omega \sim \sum_{(z,k) \in \mathcal{E}} x^z (\log x)^k \omega_{z,k}(y) \frac{dx}{x},$$

with a discrete index set $\mathcal{E}$ satisfying the usual finiteness conditions, then $\int x^w \omega$ is meromorphic in w. The b-integral is its finite part at $w = 0$. The one-term definition above is the special case needed for the trace-defect calculation; the density and the full diagonal index family must be specified in the general case.

The constant c is determined by the translation-invariant normal operator. Thus the same boundary model that obstructs trace class also determines the regularization. The finite part depends on the defining function, but the change is compensated by the trace-defect term below.

Let A be a residual b-operator whose lifted kernel has a full expansion at the front face and sufficient decay at the side faces. Its b-trace is the finite part of the diagonal integral,

$$^{b}\operatorname{Tr}(A) = \underset{z=0}{\operatorname{FP}} \int_X x^z \operatorname{tr}_E K_A(p,p)\, d\mu_b(p).$$

The ordinary trace property fails because the finite part retains a boundary contribution.

Use the Fourier convention fixed for the indicial family:

$$\widehat{f}(\lambda) = \int_{\mathbb{R}} e^{it\lambda} f(t)\, dt, \qquad f(t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-it\lambda} \widehat{f}(\lambda)\, d\lambda.$$

Theorem 17.2 (b-trace defect formula)**.** *Let A, B lie in the full b-calculus so that AB and BA are residual with the side-face decay required for the b-trace, and suppose $\partial_\lambda I(A,\lambda) I(B,\lambda)$ is trace class on ∂X and integrable in λ. Then*

$$^{b}\operatorname{Tr}([A,B]) = \frac{i}{2\pi} \int_{\mathbb{R}} \operatorname{Tr}_{\partial X} \left(\partial_\lambda I(A,\lambda)\, I(B,\lambda) \right)\, d\lambda.$$

For graded bundles, the same formula holds with the supercommutator and supertrace.

Proof. For $\operatorname{Re} z > 0$, the stated side-face orders make $x^z AB$ and $x^z BA$ trace class, so cyclicity of the ordinary trace is legitimate and gives

$$\operatorname{Tr}(x^z[A,B]) = \operatorname{Tr}([x^z, A]B).$$

Set

$$C_z = z^{-1} x^{-z} [x^z, A].$$

The full b-commutator theorem says that C_z extends holomorphically to $z = 0$ in the full calculus, and its indicial family there is

$$I(C_0, \lambda) = i\, \partial_\lambda I(A, \lambda).$$

The diagonal finite-part theorem says that the trace of $x^z C_z B$ has a simple pole at zero and identifies its residue with the Fourier integral of $I(C_0,\lambda) I(B,\lambda)$, divided by 2π. Since $[x^z, A]B = z x^z C_z B$, that residue is the constant term of the original trace. Finally $i/(2\pi) = -1/(2\pi i)$.

Holomorphic extension of C_z and the diagonal residue identification are the two analytic inputs in Melrose's b-trace theorem [3, Chapter 5]; neither follows from the displayed algebraic commutator identity alone. □

This identity is independent of the defining function: changing x changes both the finite part and the indicial family by compensating terms.

18 Dirac operators of product type

Let X be an even-dimensional compact Riemannian manifold with boundary. Assume that the metric and Clifford data are products near the boundary. Let

$$E = E^+ \oplus E^-$$

be a graded Hermitian Clifford module with compatible connection. Its total Dirac operator is odd,

$$\mathcal{D} = \begin{pmatrix} 0 & P^* \\ P & 0 \end{pmatrix}, \qquad P = D^+,$$

and near ∂X it has the form

$$\mathcal{D} = c(du)(\partial_u + \mathcal{A}), \qquad P = c(du)(\partial_u + A^+),$$

where u is the inward normal coordinate, $\mathcal{A} = \operatorname{diag}(A^+, A^-)$, and A^+ is the self-adjoint tangential operator on $E^+|_Y$.

Extend the collar coordinate u to negative values on the attached outward cylinder, put $t = -u \geq 0$, and set $x = e^{-t} = e^u$. Then $du = dx/x$ and $\partial_u = x\partial_x$, so the product formula becomes

$$\mathcal{D} = c\left(\frac{dx}{x}\right)(x\partial_x + \mathcal{A}), \qquad P = c\left(\frac{dx}{x}\right)(x\partial_x + A^+),$$

with the same tangential operators as above. This convention fixes the sign of the APS correction to be $-(\eta_{A^+}(0) + \dim\ker A^+)/2$.

The Weitzenböck formula is

$$\mathcal{D}^2 = \nabla^*\nabla + \mathcal{R},$$

where $\mathcal{R}$ is the curvature endomorphism. For the spin Dirac operator twisted by W,

$$\mathcal{R} = \frac{1}{4}\operatorname{scal} + \frac{1}{2}\sum_{j,k} c(e_j)c(e_k)F^W(e_j, e_k).$$

19 The b-heat kernel

The heat kernel is constructed on the parabolic b-heat space

$$X^2_{b,h} = \left[X^2_b \times [0,\infty)_t; \Delta_b \times \{0\}\right]_{\mathrm{par}},$$

where the final blow-up assigns weight one to spatial variables and weight two to t. The new temporal front face resolves the singularity of the kernel as $t \downarrow 0$.

At the temporal front face, the model equation is the Euclidean heat equation on each b-tangent fibre. At the boundary front face, the model is the heat equation for the normal operator on the cylinder. This local heat construction does not require full ellipticity. Full ellipticity and the boundary spectral gap enter later, when the large-time limit is identified with a Fredholm index.

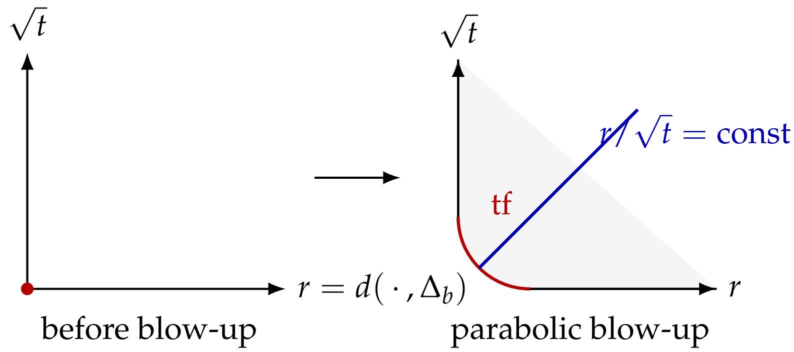


Figure 11: The temporal diagonal is parabolically blown up. The new face keeps the heat variable $r/\sqrt{t}$ finite and carries the Gaussian model.

Proposition 19.1 (Compatibility of the heat models)**.** *The Euclidean model at the temporal front face and the cylindrical normal model at the boundary front face have the same restriction to their intersection. They therefore determine a single leading kernel on $X^2_{b,h}$.*

Proof. Near a boundary point use b-normal coordinates

$$(u, y) = (-\log x, y).$$

After parabolic rescaling about the diagonal, the coefficients of $\mathcal{D}^2$ freeze at the base point. The temporal model is therefore the Gaussian heat kernel of the constant-coefficient operator on ${}^bT_pX$. If the base point tends to the boundary, the same frozen operator splits into the normal line in u and the tangential operator on T_yY. This is exactly the small-time restriction of the translation-invariant normal heat kernel. Thus both models restrict to

$$(4\pi)^{-n/2}e^{-|V|^2/4}$$

times parallel transport at the corner of the two faces. $\square$

19.1 The semiclassical construction in square-root time

Put

$$s = \sqrt{t}.$$

The parabolic blow-up in the variable t is the ordinary radial blow-up in the square-root-time smooth structure:

$$\widetilde{X}^2_{b,h} = \left[X^2_b \times [0,\infty)_s; \Delta_b \times \{0\}\right].$$

Thus the semiclassical parameter $s \downarrow 0$ and the heat-space description are two coordinate versions of the same construction. This observation is useful because it turns the heat equation into a pair of explicit model equations.

Let $n = \dim X$. Near the temporal front face, use the rescaled difference variable

$$V = \frac{z - z'}{s} \in {}^bT_zX.$$

After the density factor is removed, a heat kernel has a formal expansion of the form

$$K(s;z,z') \sim s^{-n} \sum_{j\geq 0} s^j k_j(z,V).$$

Its leading temporal symbol is

$$k_0(z,V) = (4\pi)^{-n/2} e^{-|V|_z^2/4}\, \mathrm{Id}_{E_z}\,. \tag{1}$$

Indeed, if $\Delta_V = -\sum_i \partial_{V_i}^2$ is the positive Euclidean Laplacian, then the scalar factor $G(V)$ in (1) is the uniquely normalized Schwartz solution of

$$\left[\Delta_V - \frac{1}{2}\big(V\cdot\partial_V + n\big)\right] G = 0, \qquad \int_{{}^bT_zX} G(V)\, dV = 1.$$

For the coefficient $s^{-n+j}k_j$, the corresponding normal operator is

$$T_j = \Delta_V - \frac{1}{2}\big(V\cdot\partial_V + n - j\big).$$

The equations $T_j k_j = r_j$ are the usual heat transport equations in semiclassical notation.

The second model is obtained by restricting to the boundary face and Fourier transforming in the translation variable of the cylinder. For the product operator

$$\mathcal{D} = c\left(\frac{dx}{x}\right)(x\partial_x + \mathcal{A}),$$

the indicial families satisfy

$$I(\mathcal{D},\lambda) = c\left(\frac{dx}{x}\right)(i\lambda + \mathcal{A}), \qquad I(\mathcal{D}^2,\lambda) = \lambda^2 + \mathcal{A}^2. \tag{2}$$

Consequently the boundary heat model is

$$I(e^{-t\mathcal{D}^2},\lambda) = e^{-t(\lambda^2+\mathcal{A}^2)} = e^{-t\lambda^2} e^{-t\mathcal{A}^2}. \tag{3}$$

After inverse Fourier transform, this is the one-dimensional Gaussian in the cylindrical variable multiplied by the tangential heat kernel. A different Mellin convention replaces $i\lambda$ by its signed variant; the invariant statement is $I(\mathcal{D}^2,\lambda) = \lambda^2 + \mathcal{A}^2$ on the real Fourier line.

Proposition 19.2 (Formal semiclassical heat-parametrix recursion)**.** *Let $\mathcal{D}^2$ be a product-type generalized b-Laplacian, and assume the heat-space restriction and extension theorem for polyhomogeneous kernels. Starting with the compatible leading models above, for every N one can choose a heat parametrix whose error vanishes to order N at both model faces. The compatible formal expansions can be Borel-summed to a parametrix H_0 whose error vanishes to infinite order at every temporal face.*

Proof. Assume that the lower-order jets have already been removed, and denote the first remaining temporal and boundary error coefficients by R_{tf} and $R_{\mathrm{bf}}(\lambda)$. At the temporal model, the correction with zero initial value is given by the Euclidean Duhamel operator

$$Q_{\mathrm{tf}}(t) = -\int_0^t e^{-(t-r)\Delta_{{}^bT_zX}} R_{\mathrm{tf}}(r)\,dr.$$

Thus

$$(\partial_t + \Delta_{{}^bT_zX})Q_{\mathrm{tf}} = -R_{\mathrm{tf}}.$$

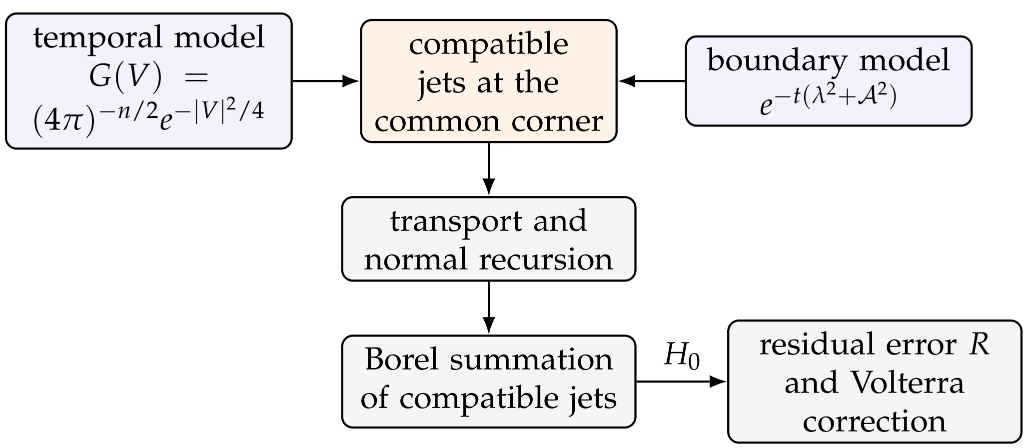
Expanding this identity in powers of s gives precisely the transport equations $T_j k_j = r_j$ displayed above.

At the boundary model, put $B(\lambda) = \lambda^2 + \mathcal{A}^2$. The corresponding correction is

$$Q_{\mathrm{bf}}(\lambda, t) = -\int_0^t e^{-(t-r)B(\lambda)} R_{\mathrm{bf}}(\lambda, r)\,dr, \qquad (\partial_t + B(\lambda))Q_{\mathrm{bf}} = -R_{\mathrm{bf}}.$$

The two Duhamel constructions commute with restriction to the common corner because the model operators agree there. Hence the two new jets can be chosen compatibly. Alternating the corrections raises the order of vanishing at each face without changing the jets already fixed at the other face: the extension theorem realizes a new face correction with zero lower-order jets at the other face.

Induction gives compatible jets to arbitrary order. The Borel extension lemma on manifolds with corners realizes those jets by a single kernel. At this point the error is residual and vanishes to infinite order at the temporal faces. This last passage uses the heat-calculus extension, composition, and push-forward results; it is the nonformal geometric input in the recursion. □



$$H = H_0 - H_0 * R + H_0 * R * R - \cdots$$

Figure 12: The temporal Gaussian and the boundary indicial heat family supply compatible model data. Formal corrections determine all face jets, Borel summation gives a global parametrix, and the Volterra inverse removes the residual error.

For a positive elliptic b-operator of order m, the same bookkeeping uses $s = t^{1/m}$. The APS problem involves $\mathcal{D}^2$, so $m = 2$ and $s = \sqrt{t}$.

Question 19.3. *Does the semiclassical construction by itself prove the APS index formula?*

Answer. No. It constructs the heat kernel and controls its asymptotic structure. The index formula still requires the local Clifford-supertrace calculation, the b-trace defect formula, regularity of the eta function at the origin, and the large-time identification with the Fredholm index. Those inputs are stated separately below. □

Matching the leading Gaussian is necessary but not sufficient: every coefficient in the two polyhomogeneous index families must agree at the corner.

Theorem 19.4 (b-heat kernel theorem; quoted)**.** *For a product-type generalized b-Laplacian, the temporal and boundary model problems have compatible complete index families on $X^2_{b,h}$. There is a heat parametrix with the prescribed normal operators whose error is residual and vanishes to infinite order at all temporal faces. Its Volterra correction is the heat kernel and remains in the b-heat calculus.*

This is the b-heat theorem of [3, Chapters 6–7]. Its proof uses the heat-space composition and push-forward theorems at every face. The factorial Volterra estimate below completes the final residual correction; it does not construct the compatible index families by itself.

The preceding semiclassical recursion describes the formal part of this theorem. The quoted heat-space results justify the global extension and the closure of the polyhomogeneous class under all the compositions used in that recursion.

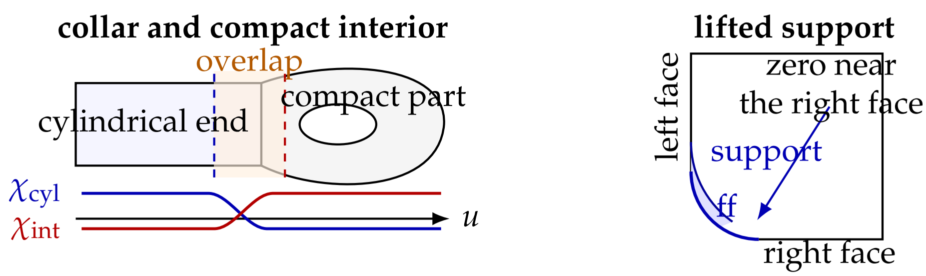


Figure 13: Left: compatible cutoffs separate the compact interior from the product cylinder while retaining an overlap on which the two model kernels can be compared. Right: after lifting to the b-double space, the boundary correction is supported near the front face and away from the opposite side face. This support control makes the alternating temporal and boundary corrections compatible.

Starting with the Gaussian model, solve the transport equations successively along the temporal front face. Borel summation gives a parametrix $H_0(t)$ satisfying the distributional initial condition $H_0(0) = I$ and

$$(\partial_t + \mathcal{D}^2)H_0(t) = R(t),$$

where R vanishes to infinite order at $t = 0$. The Volterra series

$$H = H_0 - H_0 * R + H_0 * R * R - \cdots$$

converges for small t, and the semigroup property extends it to all positive times. Here

$$(K_1 * K_2)(t) = \int_0^t K_1(t-s)K_2(s)\,ds.$$

The decisive boundary term in differentiating a Volterra convolution is

$$(\partial_t + \mathcal{D}^2)(H_0 * S) = S + R * S. \tag{4}$$

Indeed, differentiating the upper integration limit contributes $H_0(0)S(t) = S(t)$; the remaining terms give $R * S$. Applying (4) term by term makes the geometric cancellation in the displayed series explicit. Thus $H(t) = e^{-t\mathcal{D}^2}$.

Convergence comes from the volume of the ordered time simplex. On each residual heat-calculus seminorm, H_0 and R are uniformly bounded for $0 < t \leq t_0$. If $\|R(t)\| \leq C$, then

$$R^{*k}(t) = \int_{\substack{s_j \geq 0 \\ s_1+\cdots+s_k=t}} R(s_1)\cdots R(s_k)\,ds_1\cdots ds_{k-1}$$

and the simplex has volume $t^{k-1}/(k-1)!$. Hence

$$\|R^{*k}(t)\| \leq \frac{C^k t^{k-1}}{(k-1)!}.$$

The factorial proves absolute convergence; multiplication by the fixed bounded left factor H_0 gives the same conclusion for every term of the heat series. The estimate after applying b-derivatives proves convergence in the residual heat calculus. Since R vanishes to infinite order at $t = 0$, so does every correction term, and the initial value remains the identity.

The construction also proves Duhamel's formula. If L_r is a smooth family of nonnegative self-adjoint elliptic operators with common domain and is differentiable in the graph topology, then

$$\frac{d}{dr}e^{-tL_r} = -\int_0^t e^{-(t-s)L_r}\dot{L}_r e^{-sL_r}\,ds.$$

Both sides solve the same inhomogeneous heat equation with zero initial value, which proves the identity by uniqueness.

20 Small-time supertrace

For an ordinary Laplace-type operator, the diagonal heat kernel has an expansion

$$K(t;p,p) \sim (4\pi t)^{-n/2}\sum_{j=0}^{\infty} t^j a_j(p).$$

For a Dirac operator, Clifford supertrace cancels all terms below total degree n. Getzler rescaling identifies the surviving term.

Theorem 20.1 (Local index theorem)**.** *For a graded Dirac operator,*

$$\lim_{t\downarrow 0} \operatorname{Str} K(t;p,p)\, d\operatorname{vol}(p) = \left[\widehat{A}(TX,\nabla^{TX})\operatorname{ch}(E/S,\nabla^{E/S})\right]_{[n]}.$$

For the spin Dirac operator twisted by W, the second factor is $\operatorname{ch}(W,\nabla^{W})$.

The rescaled heat-kernel convergence and Clifford-supertrace normalization in this theorem are the standard local index theorem, used here as an analytic input (see [3] and the closed-manifold references cited there). The heat-space construction above supplies the boundary-compatible kernel but does not, by itself, prove the Getzler model calculation.

On a manifold with cylindrical end, diagonal integration is interpreted as a b-integral. The heat-space expansion shows

$$\lim_{t\downarrow 0} {}^{b}\operatorname{Str}(e^{-t\mathcal{D}^2}) = \int_X \left[\widehat{A}(TX)\operatorname{ch}(E/S)\right]_{[n]}.$$

Because the data are of product type, no additional local boundary density appears. This assertion is a corollary of the quoted b-heat/Getzler theorem: the temporal-front-face supertrace is the interior Getzler model, while the boundary-front-face model has zero independent constant term. It does not follow from matching only the leading Gaussian.

21 The eta invariant

For $\operatorname{Re} s$ large, define

$$\eta_{A^+}(s) = \sum_{\lambda\in\operatorname{Spec}(A^+)\setminus\{0\}} \operatorname{sign}(\lambda)|\lambda|^{-s}.$$

The Mellin representation is

$$\eta_{A^+}(s) = \frac{1}{\Gamma((s+1)/2)}\int_0^\infty t^{(s-1)/2}\operatorname{Tr}(A^+e^{-t(A^+)^2})\,dt.$$

The large-time integral is entire after omission of the zero eigenspace. The heat expansion gives a meromorphic continuation of the small-time integral.

Theorem 21.1. *The eta function of a self-adjoint elliptic differential operator B of positive order on a closed manifold is regular at $s=0$.*

Proof using the residue theorem for projections. Split the Mellin integral at $t=1$. On the orthogonal complement of the kernel, the large-time integrand is exponentially decreasing, so that part is entire. For an operator of order r on a d-manifold, the pseudodifferential heat calculus gives a polyhomogeneous expansion of $\operatorname{Tr}(Be^{-tB^2})$ as $t\downarrow 0$. Termwise Mellin integration gives the meromorphic continuation.

The standard Wodzicki–Guillemin residue theorem identifies the possible residue at zero with a constant multiple of the noncommutative residue of the order-zero operator

$$\operatorname{sgn}(B) = B|B|^{-1}$$

on $\ker B^{\perp}$. Modulo a finite-rank operator, $\operatorname{sgn}(B) = 2\Pi_+ - I$, where Π_+ is the positive spectral projection. The same theorem says that the noncommutative residue vanishes on the identity, on finite-rank operators, and on pseudodifferential projections. The possible residue is therefore zero. For a Dirac-type operator the same cancellation is visible directly in the parity of its local Clifford heat coefficients. These residue facts are quoted analytic inputs, not consequences of the preceding Mellin manipulation; see [3] for their use in the APS setting. □

For a Dirac-type operator, regularity is also visible from the parity of the Clifford heat coefficients. Write

$$\eta(A^+) = \eta_{A^+}(0), \qquad h(A^+) = \dim\ker A^+.$$

22 Heat proof of the APS index theorem

Assume first that A^+ is invertible. Set

$$F(t) = {}^b\operatorname{Str}(e^{-t\mathcal{D}^2}) = {}^b\operatorname{Tr}(e^{-tP^*P}) - {}^b\operatorname{Tr}(e^{-tPP^*}).$$

Since $\mathcal{D}$ commutes with $e^{-t\mathcal{D}^2}$,

$$F'(t) = -\frac{1}{2}\,{}^b\operatorname{Str}\left([\mathcal{D}, \mathcal{D}e^{-t\mathcal{D}^2}]_{\mathrm{s}}\right).$$

Apply the graded b-trace defect formula and use

$$I(\mathcal{D},\lambda) = c\left(\frac{dx}{x}\right)(i\lambda + \mathcal{A}).$$

Moreover,

$$I(\mathcal{D}e^{-t\mathcal{D}^2},\lambda) = c\left(\frac{dx}{x}\right)(i\lambda + \mathcal{A})e^{-t(\lambda^2+\mathcal{A}^2)}, \qquad \partial_\lambda I(\mathcal{D},\lambda) = i\,c\left(\frac{dx}{x}\right).$$

The Clifford relation $c(dx/x)^2 = -I$, together with the graded matrix trace, transports the E^--block through $c(dx/x)$ and combines it with the E^+-block. Writing $c = c(dx/x)$, the relation $cA^+ = -A^-c$ shows that the surviving contribution is the ordinary trace of $A^+e^{-t(A^+)^2}$ on $E^+|_Y$; the term odd in λ integrates to zero. Finally,

$$\int_{\mathbb{R}} e^{-t\lambda^2}\,d\lambda = \sqrt{\frac{\pi}{t}}.$$

Substitution into the defect formula, including the factor $1/2$ above, gives

$$F'(t) = -\frac{1}{2\sqrt{\pi t}}\operatorname{Tr}(A^+e^{-t(A^+)^2}).$$

This is the only place where the b-supertrace varies with t.

The fully elliptic large-time heat theorem and the spectral gap at the boundary imply that the b-heat kernel converges to the finite-rank L^2-kernel projections (whose b-traces are ordinary traces), hence

$$\lim_{t\to\infty} F(t) = \operatorname{ind}(D^+_{\mathrm{APS}}).$$

The local index theorem gives

$$\lim_{t\downarrow 0} F(t) = \int_X \left[\widehat{A}(TX)\operatorname{ch}(E/S)\right]_{[n]}.$$

Integrating the trace-defect identity and using the Mellin formula at $s = 0$ yields

$$\operatorname{ind}(D^+_{\mathrm{APS}}) = \int_X \left[\widehat{A}(TX)\operatorname{ch}(E/S)\right]_{[n]} - \frac{1}{2}\eta(A^+).$$

Indeed,

$$F(\infty) - F(0) = -\frac{1}{2\sqrt{\pi}}\int_0^\infty t^{-1/2}\operatorname{Tr}(A^+ e^{-t(A^+)^2})\,dt = -\frac{1}{2}\eta_{A^+}(0).$$

At the lower endpoint, this means: first insert $t^{s/2}$ for $\operatorname{Re} s$ large, use the Mellin identity, and then continue to $s = 0$.

If A^+ has kernel, a constant shift of its zero modes along the entire cylinder is not a finite-rank perturbation of the cylindrical operator. Use instead the global chiral b-smoothing perturbation T_b^+ constructed in the K-theoretic section. The b-heat theorem applies to $\mathcal{D}_T$, and smoothing does not change the interior local index density.

Theorem 22.1 (Ideal-boundary perturbation comparison; quoted)**.** *Let $V = \ker A^+$, choose the involution $T = -I_V$, and let T_b^+ be a global b-smoothing realization with suspended normal family*

$$I(T_b^+, \lambda) = c(dx/x)\varepsilon\rho(\lambda)\Pi_V.$$

Then $P_T = P + T_b^+$ and D^+_{APS} have naturally identified kernels and adjoint kernels. Moreover the generalized boundary correction in the b-trace-defect formula for $\mathcal{D}_T$ is

$$\eta(A^+) - \operatorname{sign}(T) = \eta(A^+) + \dim V.$$

This is Loya's ideal-boundary comparison theorem, equivalently the Melrose–Piazza spectral-section comparison [2, 5]. The sign is now explicit: $\operatorname{sign}(-I_V) = -\dim V$. Analytically, the suspended family is homotoped through invertibles to the direct sum of the nonzero family $i\lambda + A^+$ and the finite-dimensional positive zero-mode model $i\lambda + \varepsilon$; the first contributes $\eta(A^+)$ and the second contributes $\dim V$. Applying the preceding trace-defect calculation to $\mathcal{D}_T$ therefore gives $-\big(\eta(A^+) + \dim V\big)/2$. This comparison step cannot be replaced by perturbing the boundary family without constructing a global b-operator.

Theorem 22.2 (Atiyah–Patodi–Singer)**.** *Let D^+ be a product-type graded Dirac operator on an even-dimensional compact manifold with boundary. With APS boundary condition*

$$\Pi_{\geq 0}(A^+)(u|_{\partial X}) = 0,$$

one has

$$\boxed{\operatorname{ind}(D^+_{\mathrm{APS}}) = \int_X \Big[\widehat{A}(TX)\,\mathrm{ch}(E/S)\Big]_{[n]} - \frac{\eta_{A^+}(0) + \dim\ker A^+}{2}.}$$

The interior term is local. The eta invariant and the kernel dimension are the global spectral correction forced by the cylindrical end.

Question 22.3. *Why can the boundary correction not be another local curvature integral?*

Answer. The derivative $F'(t)$ is the full spectral trace $\operatorname{Tr}(A^+ e^{-t(A^+)^2})$, not the integral of one fixed heat coefficient. Integrating it from zero to infinity uses every eigenvalue of A^+. Small-time asymptotics control the meromorphic continuation, but the value $\eta_{A^+}(0)$ also contains large-time information and therefore depends on the global spectral asymmetry. This is exactly the information lost when a local index density is integrated on a manifold with boundary. □

References

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